Post-Quantum Cryptography Migration

Part 1 - Silicon & Firmware: Hardware Acceleration & Silicon

Compiled by the Spaced Research Ledger

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The new post-quantum algorithms are only useful if hardware can run them fast. This report covers that problem in three sections. Section 1 follows the tuned code that makes the new math fast on ordinary processors. Section 2 follows the dedicated accelerator cards built for heavy connection loads. Section 3 follows the ready-made circuit blocks that put the algorithms into future chips.

The clearest change this summer happened in software, not silicon. Optimized assembly implementations of ML-KEM and ML-DSA, the two workhorse post-quantum algorithms, became the default in liboqs, a widely used open-source cryptography library. wolfSSL finished dropping its outside dependency and now ships its own native, optimized implementations. Processor instruction sets themselves have not moved. Work on ARM and RISC-V instructions built for this math remains academic research, with none of it merged into products or ratified standards.

Dedicated hardware is arriving for the heaviest loads. FPGA-based accelerator cards from Silicom and Eideticom now handle millions of post-quantum TLS handshakes per second. NVIDIA GPUs speed up key encapsulation about tenfold through the cuPQC library. Intel has promised full post-quantum compliance across new platforms by 2030. Whether its QuickAssist accelerators already speed up the new algorithms is disputed, and Section 2 states that disagreement in full.

For chips themselves, licensable design blocks from Rambus, Synopsys, PQShield, and PQSecure now cover the standardized algorithm set. Validated end products are starting to follow. Crypto4A's hardware security module is described as the first quantum-safe module to pass FIPS 140-3 Level 3 validation, in August 2026. STMicroelectronics put a post-quantum accelerator into a mobile security chip that also carries NFC and eSIM functions on one die.

1. CPU & ISA Extensions

Every secure connection a computer makes runs cryptographic math on its main processor. This section covers how fast the new post-quantum algorithms run on mainstream chips from Intel, ARM licensees, and RISC-V designers. Speed can come from two places: hand-tuned code that uses the vector instructions chips already have, and proposals for new instructions built for the algorithms' distinctive math. Today all the practical gains come from tuned code inside cryptographic libraries. The new instructions remain research.

Recent Developments

The news this summer is that optimized implementations became the ones everyone gets by default. liboqs, the open-source library many products build on, made tuned assembly standard in July. Its 0.16.0 release brings in mldsa-native and updates mlkem-native to version 1.2.0, with optimized x86-64 and ARM builds enabled out of the box [1]. wolfSSL went further in August and removed its dependency on liboqs entirely. It now ships its own native implementations of ML-KEM and ML-DSA with optimized paths for both major architectures [2].

Current State

Everything fast today comes from software making better use of instructions that chips already have. liboqs had already adopted mlkem-native, with portable C, AVX2, and ARM variants, as its standard ML-KEM implementation [3]. The first stable version of that code arrived with liboqs 0.14.0 in August 2025, alongside a faster SHA-3 built on AVX-512VL instructions [4].

The optimized variants come straight from the actively maintained upstream mlkem-native project [5], which recently added a consistency self-test and support for Intel's control-flow protection [1]. Much of the code is formally verified: the C for memory and type safety, the ARM assembly for functional correctness [1]. When built for distribution, the library picks the fastest routine for the running processor automatically [6][7]. Only the standardized algorithm names are treated as stable identifiers, and everything else may still change [8].

Chipmakers contribute tuning rather than new hardware. Intel published AVX-512 optimizations of the library's Keccak hashing core, worth up to a 1.64x speedup on Xeon processors [9]. An experimental AVX-512 build of ML-KEM, running 32 polynomial multiplications in parallel, has been wired into liboqs and an OpenSSL provider [10]. NVIDIA's GPU library cuPQC was added as a further backend [1]. wolfSSL maintains its own AVX2 assembly for ML-KEM and fixed bugs in it during its 5.9.1 release cycle [11].

The craft involved is considerable. State-of-the-art implementations for small ARM microcontrollers rely on more than 3,000 lines of hand-written assembly per algorithm [12]. A tool called SLOTHY can now generate code that matches or beats hand-optimized assembly for ARM's Neon and Helium instruction sets [12].

Earlier NEON work covered the pre-standardization candidates on hardware from the Apple M1 down to the Raspberry Pi 4 [13]. The formally verified Libjade library shows the cost of full rigor. It still offers only two Kyber variants, on x86-64 Linux and macOS alone [6].

New instructions, by contrast, are still in the lab. On ARM, the research is real but unadopted. ARMv9 processors carry the SVE2 and SME vector extensions, with SVE2 folding the older NEON instructions into a broader superset [14].

A January 2026 paper used both to run ML-KEM up to 7.77x faster than reference code on an Apple M4 Pro, the first such work on real ARMv9 hardware [15]. A companion effort made ML-DSA signing 70.7 percent faster using SVE2 [16]. The catch is unchanged: none of these code paths have been merged into production libraries such as liboqs, OpenSSL, or mlkem-native [15].

RISC-V is further behind. No ratified RISC-V crypto extension includes instructions for lattice math, and every acceleration effort found remains an academic proposal [17]. The proposals are serious designs. RISQ-V couples 29 custom instructions to the processor pipeline and shows up to 11.4x speedups on prototype hardware [17]. A fabricated test chip with a tightly integrated NTT accelerator improved end-to-end ML-KEM performance by up to 56.5 percent [18].

Others include the HORCRUX extension architecture [19], the VPQC vector processor [20], a programmable crypto-processor covering the NIST algorithms [21], and simulator studies of vector-extension speedups for

Classic McEliece [22]. All of them stop short of the step that would matter to products: adoption into a ratified extension [17].

2. Offload & Accelerators

Offload hardware takes cryptographic work off a server's main processor and runs it on a dedicated card or chip. The devices that terminate huge numbers of encrypted connections, such as load balancers and data-center network cards, need this the most. Post-quantum handshakes cost more compute than the ones they replace, so at millions of connections per second the math earns its own silicon. This section covers the accelerator cards, SmartNICs, and GPU libraries being adapted for the new algorithms.

Recent Developments

August brought hard performance numbers for dedicated cards, and a long-range pledge from Intel. Silicom's Accelerated Crypto Adapter, built on Eideticom's NoLoad technology, handles up to 450,000 hybrid post-quantum handshakes per second. The vendors equate that to offloading more than 85 processor cores [23]. Silicom quotes 1.8 million TLS handshakes per second for ML-KEM-768 on a related FPGA design [24]. Intel, for its part, has committed to full post-quantum compliance across new platforms by 2030 [25].

Current State

Most of the market handles post-quantum handshakes without any dedicated card at all. The big traffic-management vendors added them in plain software. F5's BIG-IP runs hybrid ML-KEM key exchange on its general compute, and F5 states that no hardware security module or dedicated crypto silicon is required [26].

Support has widened release by release. BIG-IP Next 20.3 carried a draft hybrid cipher [27], version 17.5 added both client and server sides [28], and the NIST-approved X25519 combination followed [29]. Version 21.1 added two more NIST-compliant cipher groups this May [30].

Citrix NetScaler supports hybrid key exchange on the front end [31]. A10 went one step further and enabled it by default on client-side connections through OpenSSL 3.5 [32].

Where dedicated hardware does appear, it comes in three forms. Market analysis describes FPGA, ASIC, and system-on-chip accelerators as the dominant product types [33]. Eideticom's accelerator card offloads signature generation and verification and plugs into OpenSSL and NGINX [34]. NVIDIA's cuPQC library, being folded into liboqs through a Linux Foundation partnership [33], runs ML-KEM up to 10x faster than CPUs. It reaches 3.16 million key generations per second for the smallest variant [35].

On the network side, BlueField data-processing units have carried post-quantum IPsec at 400 gigabits per second in demonstrations [36]. That capability comes from a third-party integration, Qrypt's gateway software, rather than anything NVIDIA-native [37]. Qrypt has extended the same gateway to the new 800-gigabit BlueField-4 [38], and AIC ships a FIPS-compliant server platform pairing BlueField-3 with Qrypt's encryption [39]. Bloombase integrates its storage encryption with Intel's Mount Evans network processor [40].

Intel's present-day contribution is narrower than its 2030 pledge suggests. The company says it began adding post-quantum capabilities in 2023 [25], but an independent evaluation found its QuickAssist accelerator a poor fit for hash-based signatures. Those schemes issue many small hash calls, where the accelerator only wins on large inputs [41].

Unresolved Conflicts

One real disagreement stands, over Intel's QuickAssist. Intel's own technical material shows its acceleration paths, the IPsec Multi-Buffer library and the Cryptography Primitives Library, speeding up classical algorithms rather than offloading the new ones in hardware [42]. Its demonstrated post-quantum gains come from AVX-512 instructions on Xeon CPUs, not from the QuickAssist accelerator [43][44].

Against that, NEXCOM markets an edge server that it says uses QuickAssist to accelerate post-quantum workloads [45]. A specification showing ML-KEM or ML-DSA offload running on QuickAssist hardware would settle the question. Nothing public does so today.

3. PQC Silicon IP Cores

Most chipmakers do not build cryptography from scratch. They license ready-made circuit designs, called IP cores, and drop them into their own processors, secure elements, and hardware security modules. A chip fixed in silicon cannot be patched the way software can, so chips being designed today must already include the new algorithms. This section covers the vendors selling those building blocks and the first finished chips that carry them.

Recent Developments

Validated post-quantum silicon stopped being theoretical this summer. Crypto4A's QASM module is described as the first quantum-safe HSM to pass FIPS 140-3 Level 3 validation, in August 2026 [46]. Thales launched Luna 8, new hardware with a custom cryptographic processor, now under the same FIPS and Common Criteria evaluations [47]. PQShield's PQCryptoLib-Core 3.5.1 passed NIST validation covering ML-KEM, ML-DSA, and the SHA-3 family in August [48]. PQSecure's crypto-agile framework passed NIST validation in June [49].

New chips and designs followed the validations. STMicroelectronics announced the ST54M, a mobile chip combining a post-quantum accelerator with NFC, a secure element, and an eSIM on a single die [50]. Rambus introduced the RT-648, an ARM-based embedded security module for automotive silicon [51]. PQSecure demonstrated its core running in QuickLogic's embedded FPGA fabric on Intel's 18A process, a trade study rather than a shipping chip [52][53].

One timeline moved the other way. SEALSQ's Common Criteria milestones slipped from March to September 2026, with the full post-quantum test report now expected in December [54].

Current State

Existing secure hardware mostly gains the algorithms through firmware, not replacement. Vendors of hardware security modules ship the algorithms as firmware updates, usually with hybrid classical-plus-post-quantum modes for cautious pilots [55]. Thales Luna firmware 7.9.0 brought validated ML-KEM and ML-DSA, with the LMS scheme supported from 7.8.9 [56]. Entrust's nShield 5 firmware carries algorithm validations for ML-DSA, ML-KEM, and SLH-DSA [57]. Eviden, Fortanix, and Futurex all report supporting the full NIST algorithm trio [58].

Two structural drags apply across the board. The FIPS validation queue adds roughly 12 to 18 months between an algorithm existing and a certified product carrying it [59]. And the updated TPM 2.0 specification now defines post-quantum identifiers, but deployed TPM chips are usually soldered down and only gain the algorithms if manufacturers ship firmware updates [59].

Four vendors sell the building blocks for new chips. Rambus offers the broadest stack. Its CryptoManager root-of-trust family has quantum-safe variants [60], and the QSE-IP-86 engine accelerates all three NIST algorithms with a dedicated lattice-math core [61]. The standalone Quantum Safe Engine has been on offer since 2023 [62], alongside an FPGA security suite with side-channel countermeasures [63] and pre-validated automotive HSM software built with ETAS [64]. The platform pitch throughout is agility: algorithm updates by firmware instead of silicon redesign [65].

Synopsys takes the same hybrid hardware-and-firmware approach in its Agile PQC public-key accelerators, which cover the NIST algorithms plus the hash-based XMSS and LMS schemes [66][67]. The accelerators handle signatures, key encapsulation, and key generation, with FIPS 140-3 support and optional side-channel countermeasures [68]. They sit inside a broader security portfolio positioned for chips designed ahead of the standards [69]. Its physical-unclonable-function key storage rests on symmetric cryptography and needs no post-quantum retrofit at all [70].

PQShield pairs certified software with silicon designs. Its embedded library fits the algorithms into under 5 kilobytes of RAM [71] and became an ST-authorized product for STM32 microcontrollers [72]. It also supports the PSA Certified crypto API [73] and gained software-level protections against power analysis and fault injection [74].

On the hardware side, PQShield has a silicon test chip carrying its platform IP [75], including an accelerator aimed at hash-based signatures [76]. The wider product line spans acceleration engines and full root-of-trust subsystems [77]. The company says customers including Microchip have its IP in silicon today, naming AMD, Lattice, and Mirise as further collaborations. No shipping part numbers are public, so that remains a vendor claim [78].

PQSecure, the smallest of the four, sells hardware cores proven across several FPGA vendors' fabrics [79] and partners on the NIST crypto-agility project [80].

In secure elements, vendor claims still outrun independent confirmation. SEALSQ's QS7001 implements ML-KEM and ML-DSA and passed entropy-source validation in May [81]. The company describes the part as shipping today and cites a pipeline of over 150 customers, figures that are self-reported and unverified [82].

Two cautions round out the picture. On a hardware datasheet, “quantum-safe” means the firmware can run the new algorithms inside the existing tamper-resistant boundary, not that the old hardware was quantum-vulnerable [83]. And migration guides tell buyers to confirm that a vendor’s post-quantum firmware is on the roadmap and covered by the existing maintenance contract [84].

About This Report

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